The variational principle

In the first lecture, we have seen that the energy of a wavefunction that solves the SE can be obtained by calculating the expectation value with the Hamiltonian. Of course, we can calculate the expectation value for any wavefunction (including those that are not solutions of the SE). This defines a functional $F[\psi]$ according to

$$F[\psi] = \int dV \psi^*(\mathbf{r}) \hat{H} \psi(\mathbf{r}). \tag{1}$$

Note that a functional takes a function as input and gives a number as output (in contrast, a function takes a number as input and produces a number as output). One can show that

$$F[\psi] \geq \epsilon_0, \tag{2}$$

for any normalized wavefunction ψ. Here, ϵ_0 denotes the ground state energy of the Hamiltonian. In other words, the ground state energy is the minimum of the functional $F[\psi]$.
To find the argument where a function takes its minimum value, we have to set the first derivative to zero. Similarly, for the case of a functional we have to set the so-called functional derivative (denoted by $\delta/\delta\psi$) to zero. Importantly, we are only allowed to search the minimum for normalized wavefunctions. This constraint can be incorporated using the method of Lagrange multipliers. Therefore, the equation we have to solve to find the ground state wavefunction is

$$0 = \frac{\delta}{\delta\psi^*(\mathbf{r})} \left[F[\psi] - \lambda \left\{ \int dV' |\psi(\mathbf{r}')|^2 - 1 \right\} \right], \tag{3}$$

where λ is the Lagrange multiplier that enforces the normalization constraint.
To evaluate this functional derivative, you only need to know that $\delta\psi^*(\mathbf{r}')/\delta\psi^*(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}')$ and that $\delta\psi(\mathbf{r}')/\delta\psi^*(\mathbf{r}) = 0$. Using the definition of $F[\psi]$, we then find

$$0 = \int dV' \delta(\mathbf{r} - \mathbf{r}') \left[\hat{H}\psi(\mathbf{r}') - \lambda\psi(\mathbf{r}') \right] = \hat{H}\psi(\mathbf{r}) - \lambda\psi(\mathbf{r}). \tag{4}$$

Rearranging this yields

$$\hat{H}\psi(\mathbf{r}) = \lambda\psi(\mathbf{r}). \tag{5}$$

You can see that we have derived the SE from the variational principle and that the Lagrange multiplier λ plays the role of the electron energy ϵ.

Many electrons

So far, we have always solved the SE for a single electron. This was possible because electron-electron interactions were neglected. To describe systems with many electrons that all interact with each other, there are two possibilities: i) we can try to describe all electrons at the same time or ii) we can try to describe one electron at a time. Route i) is the standard formulation of quantum mechanics of many interacting electrons. Route ii) will lead us to DFT.
In the standard approach, the state of N interacting electrons is described by the many-electron wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N)$ which has the following statistical interpretation: $|\Psi(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N)|^2 dV_1 dV_2 ... dV_N$ is the probability of finding electron 1 in the volume element dV_1 at position $\mathbf{r_1}$, electron 2 in the volume element dV_2 at position $\mathbf{r_2}$ and so on. The many-electron wavefunction is obtained by solving the many-electron SE

$$\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N). \tag{6}$$

The Hamiltonian operator of the interacting many-electron system is given by

$$\hat{H} = -\sum_{i=1}^{N} \frac{\hbar^2 \nabla^2}{2m} + \sum_{i=1}^{N} V(\mathbf{r}_i) + \frac{1}{8\pi\epsilon_0} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{8\pi\epsilon_0} \sum_{I \neq J} \frac{Z_I Z_J e^2}{|\mathbf{R}_i - \mathbf{R}_j|}. \tag{7}$$

Here, the first term describes the kinetic energy of all electrons and the second term is the potential energy due to the interaction with the atomic nuclei $V(\mathbf{r}) = -\sum_I Z_I e^2/(4\pi\epsilon_0|\mathbf{r} - \mathbf{R}_i|)$ (with Z_I denoting the number of protons in the I-th atomic nucleus located at position $\mathbf{R}_I$). The third term describes the repulsive interaction between electrons. The last term captures the repulsion between the nuclei. Note that we will often assume the nuclei to be fixed in space. In that case, the last term is just a constant that can be omitted (this is also the reason why we have neglected the kinetic energy of the nuclei - we will describe the motion of nuclei and atoms later in the course).
Alternatively to solving the many-electron SE, we can also determine the ground state energy from the variational principle

$$F[\Psi] = \int dV_1 dV_2 ... dV_N \Psi^*(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N) \hat{H} \Psi(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N) \geq E_0, \tag{8}$$

with E_0 denoting the ground state energy.
It is maybe an irony of nature that it is so easy to write down the fundamental equation for interacting many-electron systems, but solving this equation for a realistic material is incredibly difficult. The reason is that the number of electrons in any material is incredibly large (on the order of the Avogadro number, i.e. 10^{23} electrons). This means that the many-electron wavefunction depends on 10^{23} position variables. Even if it was possible to compute this function, it would not be possible to store this wavefunction anyway because it contains so much information.
To make progress, we need a better way to describe interacting many-electron systems, like materials. Specifically, we do not need all of the information that is contained in the many-electron wavefunction. We are mostly interested in the ground state energy E_0 from which many experimentally important quantities, such as forces, structures and phase diagrams, can be derived.

Hohenberg-Kohn theorem

DFT is a much more useful approach for calculating the ground state energy than solving the many-electron SE. In particular, we will see that it is possible to calculate the ground state energy E_0 from the electron density $n(\mathbf{r})$. The electron density only depends on a single position variable and is therefore a much more "user friendly" quantity than the many-electron wavefunction.
The fundamental theorem of DFT is the Hohenberg-Kohn theorem which states that the potential $V(\mathbf{r})$ which enters the many-electron SE is uniquely determined by the corresponding ground state density $n(\mathbf{r})$. In other words, the theorem states that there is a one-to-one mapping between potentials and densities. This, however, also implies that the ground state density also uniquely determines all other properties of the material, including the many-electron wavefunction Ψ and the ground state energy E_0 (since $V(\mathbf{r})$ determines Ψ and E_0 through the many-electron SE). For a proof of the Hohenberg-Kohn theorem, you can read pages 38 and 39 in the Giustino book.
We next apply the Hohenberg-Kohn theorem to the variational principle

$$E_0 \leq F[\Psi] = F[\psi[n]] \equiv \mathcal{F}[n]. \tag{9}$$

Here, we used the fact that the many-electron wavefunction is determined by the ground state density, i.e. is a functional of n, and also defined the new functional $\mathcal{F}[n]$ which shows that the ground state energy E_0 can indeed be determined directly from the ground state density by minimizing the functional. Unfortunately, the Hohenberg-Kohn theorem only states that the many-electron wavefunction can be determined from the density, but it does not say how. We therefore do not know how the functional $\Psi[n]$ looks like (we only know that it must exist) and therefore we do not know how the functional $\mathcal{F}[n]$ (but we know it must exist).

Kohn-Sham approach

These shortcomings of the Hohenberg-Kohn theorem are overcome by the Kohn-Sham approach which turns DFT into a useful theory for modelling materials. Kohn and Sham noted that for a given ground state density $n(\mathbf{r})$ some contributions to the corresponding ground state energy are easy to estimate. In particular, the interaction energy with the potential of the nuclei is given by $\int dV n(\mathbf{r}) V(\mathbf{r})$. In addition, the classical electrostatic repulsion of the electrons is captured by the so-called Hartree energy

$$\frac{e^2}{8\pi\epsilon_0} \int dV \int dV' \frac{n(\mathbf{r}) n(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|}. \tag{10}$$

What is missing so far are terms that capture the kinetic energy of the electrons as well as terms describing quantum-mechanical interaction effects, such as the exchange interaction that arises from the Pauli principle.

To make progress, Kohn and Sham proposed to approximate the kinetic energy of interacting electrons by the kinetic energy of a fictitious set of non-interacting electrons (the so-called Kohn-Sham electrons) which have the same density $n(\mathbf{r})$. Since the Kohn-Sham electrons are non-interacting, they can be described by a set of one-electron wavefunctions $\{\psi_i(\mathbf{r})\}$ which are called the Kohn-Sham (KS) orbitals. The ground state density can be expressed in terms of these orbitals via

$$n(\mathbf{r}) = \sum_{i=1}^{N} |\psi_i(\mathbf{r})|^2 \tag{11}$$

and the kinetic energy of the KS electrons is given by

$$-\sum_{i=1}^{N} \int dV \psi_i^*(\mathbf{r}) \frac{\hbar^2 \nabla^2}{2m} \psi_i(\mathbf{r}). \tag{12}$$

Finally, Kohn and Sham defined the so-called exchange-correlation energy E_{xc} which contains the difference between the kinetic energy of the interacting electrons and the KS electrons as well as quantum-mechanical interaction terms, like the exchange energy. In the decades since the KS approach was proposed, much work has been done to develop approximations to E_{xc}. However, the exact expression for this term still remains unknown.

Now, that we have expressed the functional $\mathcal{F}$ in terms of four contributions (interaction energy with the nuclei, Hartree term, kinetic energy of KS electrons and exchange-correlation energy), we can use the variational principle to determine the KS wavefunctions which minimize $\mathcal{F}$. The minimum value of $\mathcal{F}$ is of course the ground state energy. Again, we include normalization constraints on the KS wavefunctions through Lagrange multipliers. This leads to

$$0 = \frac{\delta}{\delta \psi^*(\mathbf{r})} \left[\mathcal{F}[\psi] - \sum_i \lambda_i \left\{ \int dV' |\psi_i(\mathbf{r}')|^2 - 1 \right\} \right]. \tag{13}$$

Next, we have to evaluate the functional derivatives of all 4 contributions to $\mathcal{F}$. This leads to

$$0 = -\frac{\hbar^2 \nabla^2}{2m} \psi_i(\mathbf{r}) + V(\mathbf{r}) \psi_i(\mathbf{r}) + V_H(\mathbf{r}) \psi_i(\mathbf{r}) + V_{xc}(\mathbf{r}) \psi_i(\mathbf{r}) - \lambda_i \psi_i(\mathbf{r}), \tag{14}$$

where we defined the Hartree potential $V_H(\mathbf{r}) = e^2/(4\pi\epsilon_0) \int dV' n(\mathbf{r}')/|\mathbf{r} - \mathbf{r}'|$ and the exchange-correlation potential $V_{xc}(\mathbf{r}) = \delta E_{xc}/\delta n(\mathbf{r})$.

Rearranging this expression yields the KS equations

$$-\frac{\hbar^2 \nabla^2}{2m} \psi_i(\mathbf{r}) + V_{KS}(\mathbf{r}) \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r}). \tag{15}$$

Here, we defined the KS potential $V_{KS} = V + V_H + V_{xc}$ and again interpreted the Lagrange multiplier as an energy eigenvalue ϵ_i. These energies are known as the KS energies. Like the KS electrons and the KS wavefunctions, it is important to remember that these are non-physical quantities and one must be very careful in interpreting them.

Solving the KS equations for the KS wavefunctions is difficult because the KS potential depends on the density, but the density depends on the KS wavefunctions. To make progress, we can start with an initial guess for the density and construct the KS potential for this density. Then, we can solve the KS equations and calculate a new density from the KS wavefunctions. If the new density is different from the initial one, we repeat this procedure until self-consistency is achieved.